ReEQ – FMA Edition

User Manual

Parametric graphic equalizer for REAPER (JSFX) · Version 1.2.0-fma.20

1. What it is / credits

ReEQ – FMA Edition is a parametric equalizer with up to 16 bands that is drawn and driven directly on the spectrum. It is the corrected and extended edition of ReEQ v1.2.0, created by Justin Johnson (“Sai'ke”), one of the most admired equalizer designs in the JSFX world. This edition completes the work the original left pending — per-band dynamics wired up, a rebuilt HQ mode, a finished limiter and a family of visual themes — as a tribute to its great design.

This manual covers the buttons and the basics of how it operates. The rest is intuitive: if you can use a graphic equalizer, you already know almost all of ReEQ.

> **Original author:** Justin Johnson (“Sai'ke”) — design, code and interface.
>
> **FMA Edition:** Juan de Dios Vázquez Reyes — Free Mix Audio (FMA Series): audit, fixes and additions.
>
> **License:** MIT (see the LICENSE file). The audio is identical in every visual theme.

2. Installation

- In REAPER: Options → Show REAPER resource path in explorer/finder.
- Open the Effects folder and copy the contents of this package's Effects folder into it, keeping the subfolders (reeq-fma-edition-deps and Dependencies).
- In the FX browser press F5 to rescan, then insert “JS: ReEQ - Parametric Graphic Equalizer (FMA Edition)”.

Two files, one plugin — which one to install

- ReEQ.jsfx — the DROP-IN REPLACEMENT for the original: it carries Sai'ke's ReEQ file name, so if the original is already installed, just copy this file over it (or delete the old one and keep this). Your projects will load it where they used to load the original, with every setting preserved, and it opens with Sai'ke's original look (the Classic theme). You can delete the original without fear.
- reeq-fma-edition.jsfx — the edition under its own name: it opens with the FMA house theme, FMA Solid. It is exactly the same plugin and the same sound; the ONLY difference between the two files is the look they open with.
- You may install both at once: REAPER tells them apart by file name and they never conflict. Whichever you install, ALWAYS copy the include subfolders too (reeq-fma-edition-deps and Dependencies): both files share them, and without them the plugin will not load.
- Nothing is lost in the replacement: the control map is the original's, untouched; what is new in this edition (the themes, SILK) starts at its default values.
- Frequencies are preserved EXACTLY, in Hz: when an old project opens, a band's internal position number may be re-expressed once (this edition computes the scale from your session's real sample rate instead of a fixed value), but the resulting frequency is identical — as are gains, Q and slopes. The sound of your bands does not change.
- Two honest sound notes for old projects: (1) a project saved in HQ mode will play through this edition's new HQ (×2 minimum-phase oversampling — in the original, at 88.2 kHz and above, HQ silently did not apply); (2) if you used the LIMIT button with a hot signal, the ceiling is now soft and sits at −1 dBFS, instead of the original's hard clip at 0 dBFS. In ECO and without LIMIT, the sound

is the original's, with its corrections applied.

- Safe replacement, step by step: 1) copy the include subfolders into your Effects folder; 2) copy ReEQ.jsfx over the original; 3) open one of your projects and check the plugin loads with your settings; 4) from then on the old file is no longer needed — delete it whenever you like. NEW instances of the plugin start in ECO mode by design (light on CPU); the mode saved in each project overrides that default.

3. Overview

- The graph: the white curve is the total response of all bands; behind it you see the live spectrum and each band's curve in its own colour.
- The nodes: double-click an empty spot on the graph to create a band there (up to 16). Dragging a node moves frequency (horizontal) and gain (vertical). The mouse wheel over a node adjusts Q; Ctrl + wheel adjusts the slope on cuts. Double-clicking a node switches it off or on. Dragging over an empty area selects several bands at once.
- Right-click a node: a menu with Enable/Disable, Invert Gain, Zero Gain, Dynamic, Shape (filter type), Slope, Stereo Placement (group), Split, Duplicate and Delete.
- The node panel (chapter 5) appears when a band is selected, if the PANEL button is on.
- The bottom bar (chapter 4) holds the global controls, centred under the graph. Settings (top right) opens the analyzer menu (chapter 6). The dB numbers on the graph's right edge are also a button: they set the vertical range (6, 12, 18, 24 or 30 dB).

4. The bottom bar, button by button

Number buttons are adjusted by dragging vertically; double-clicking a button returns it to its default value.

ECO / HQ — the processing quality. ECO runs at the session sample rate, with zero latency (PDC 0) and minimal CPU: it is the mixing mode, and a fresh instance always opens in ECO. HQ oversamples ×2 with minimum phase: the filters run at twice the rate, with 31 samples of latency that REAPER compensates automatically (PDC 31), and it is the mode that enables SILK. Click it to open the menu and choose.

M/S — the global stereo mode: Mid/Side or Left/Right. It defines what the M: and S: trims in the bar and each band's group mean.

100 % (Scale) — scales the gain of ALL bands at once (0–200 %). At 50 %, a +6 dB boost sounds like +3 dB, without touching a single node.

0.0 dB (Gain) — master output gain.

M: 0.0 dB and Ø — independent trim for the Mid channel (or Left, depending on the stereo mode). The Ø button flips its polarity.

S: 0.0 dB and Ø — the same for the Side channel (or Right).

LIMIT — a soft output ceiling at −1 dBFS. Below −3 dBFS the signal passes untouched; above it, a smooth curve keeps it always under the ceiling, with none of the original's hard clipping at 0 dBFS. No latency in either mode.

PANEL — shows or hides the floating panel of the selected node.

AGC — automatic gain compensation: while audio plays, it compares the input and output levels and compensates the difference, so you can judge the EQ without level fooling you. The tiny “S” key glued to its right freezes the current compensation: it turns it into fixed gain (added to the master Gain) and switches AGC off.

SILK — silk amount (0–100 %), HQ only. It is a spectral exciter stage inside the oversampled path: it does not move the EQ curve (steady tones keep their level) and only adds harmonics with a character per region — deep lows, defined mids and expensive highs. At 0 % the stage disappears entirely. In ECO it is shown dimmed with a diagonal bar: it is locked, because the stage lives in the ×2 path. Drag to adjust; Ctrl + double-click returns it to 0.

GUI: — the theme selector. Clicking it opens the list upward: FMA Solid (the house base theme), FMA Fire Storm, FMA Ice Storm, FMA Sand Storm, FMA Star Rain, FMA Sea Mist, FMA Bronze Wall and Classic (Sai'ke's original look). Changing theme never alters the audio. Which look is the factory default depends on the installed file: ReEQ.jsfx opens in Classic and reeq-fma-edition.jsfx in FMA Solid (chapter 2).

5. The node panel

“< N >” — band selector: jump between the existing bands.

BYPASS — switches the selected band off without deleting it (the same as double-clicking its node).

Filter type — a pill showing the current type; click to change it among the 14: Peak, Low Cut, Low Cut (Butterworth), Low Cut (Analog), Low Shelf, High Shelf, High Cut, High Cut (Butterworth), High Cut (Analog), Notch, Band Pass, Tilt Shelf, Pultec Low Shelf and All Pass.

Group — Stereo, Mid, Side, Left or Right: which part of the stereo signal the band affects, according to the bar's M/S mode.

Slope — only appears on cuts (Low Cut / High Cut): 6, 12, 18, 24, 30, 36, 48, 72, 96 or 120 dB per octave.

FREQ / GAIN / Q dials — frequency, gain (±18 dB) and band width. Adjust them by dragging.

Per-band dynamics — the DYN dial sets the dynamic range (−24…+24 dB), THR the threshold (−60…0 dB), ATK the attack (0.5–250 ms) and REL the release (5–1000 ms). The DYN ON pill arms the band's dynamics: when the level at the band's frequency rises above the threshold, its gain moves toward the DYN value (positive = boosts, negative = cuts). On the graph it shows as a translucent fill and a live curve around the node, and the currently applied dynamic gain appears under the DYN dial. With DYN at 0 the band is static.

LISTEN — hear only this band: it isolates its region, to hunt resonances or check what the filter is doing.

DELETE — removes the selected band.

6. Settings

The Settings button (top right) opens the spectrum analyzer menu:

- Spectrum: what is drawn — Full, Mid, Side, Mid + Side, Left, Right or Left + Right.
- Display: Fill, Line or None (no spectrum).
- Range: spectrum ceiling (0, 20 or 40 dB) and floor (−90, −140 or −200 dB).
- Tilt: display tilt (0 to 6 dB/oct in 1.5 steps), to compensate music's natural fall toward the highs.
- Window Type: the FFT window — Hamming, Blackman-Harris, Blackman or Rectangular.
- FFT Size: 2048, 4096, 8192 or 16384. Larger = more low-end detail, more drawing CPU.
- Display Pre-EQ: also shows the spectrum before the EQ.

In addition, “Show Piano” and “Show Peaks” exist as effect parameters in REAPER's parameter list: the piano strip under the graph and the spectrum's peak hold.

7. Quick tips and a CPU note

- Mix in ECO; switch to HQ for mastering, or whenever you want SILK.
- SILK with moderation: 15 to 40 % is usually enough; its work shows when you compare, not instantly.
- To tame a harshness that only shows on peaks: a Peak band with negative DYN, the threshold just under the peaks and a short REL.
- Turn LIMIT on if your chain can pass 0 dBFS: the −1 dBFS ceiling leaves headroom for conversion to lossy formats.

CPU note: ECO is light (a single filter cascade at the session rate). HQ costs roughly twice as much in the filters, plus the half-band FIR; SILK adds a little more, in HQ only.

ReEQ was created by Justin Johnson (“Sai'ke”) and shared under the MIT license. This FMA edition is a tribute to its design.